

**SERVICENOW**

# **MEDICAL DIAGNOSTIC CENTER**

*Project Report*

**Submitted by**

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ServiceNow Global Certification Project

Patient Management | Appointment Scheduling | Report Generation

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# 1. INTRODUCTION

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The Medical Diagnostic Center application built on ServiceNow automates and streamlines patient registration, diagnostic test catalog management, appointment booking, report generation, email notifications, and health analytics. In a modern healthcare environment, manual record-keeping leads to delays in diagnostic reporting, scheduling conflicts, and communication gaps between healthcare providers and patients.

This project demonstrates a custom ServiceNow application designed to manage the entire lifecycle of a patient's interaction with a diagnostic laboratory. From cataloging laboratory tests (such as Blood Tests, COVID-19 PCR, and Imaging) to generating diagnostic reports and dispatching automated email notifications, the solution provides an end-to-end, traceable, and secure workflow.

## Project Overview

- **Platform:** ServiceNow Custom Application Development
- **Primary Module:** Medical Diagnostic Management
- **Supporting Modules:** Platform Analytics, Email Notification Engine, System Logs
- **Primary Roles:** Patient / End-User, Lab Technician, Medical Approver / Admin, System Administrator
- **Main Activities:** Test catalog definition, patient onboarding, appointment booking, automated billing & approval routing, report generation, email notification triggering, and reporting dashboard analytics
- **Key Entities:** Patient Table, Diagnostic Test Table, Appointment Table, Report Table, Email Notifications, Health Reports Dashboard
- **Outcome:** A centralized, automated, and secure healthcare diagnostic management portal
- **Overall Benefit:** Faster report turnaround times, automated patient notifications, zero double-booking, transparent billing, and enhanced operational visibility via analytics dashboards

## 2. PROJECT OBJECTIVES AND SCOPE

### 2.1 Business Objectives

- Standardize diagnostic test cataloging with dynamic pricing, durations, and test codes.
- Streamline patient registration while securely storing personal health data.
- Provide a self-service portal for patients to search tests, schedule appointments, and receive confirmation notifications.
- Automate the approval workflow for test appointments, payment processing, and report delivery.
- Enable lab technicians to upload findings and automated email notifications with dynamic field scripts (`${patient.first_name}`, `${report_number}`).
- Deliver real-time analytics to clinic managers via custom Platform Analytics dashboards.

### 2.2 Functional Scope

- Custom table creation: Diagnostic Test, Patient, Appointment Table, and Report Table.
- Field-level configurations including reference fields, currency fields, choice dropdowns, HTML editors, and date-time pickers.
- Dynamic self-service portal UI for patient appointment scheduling and live confirmation tracking.
- Automated email notifications for appointment status and report availability.
- Custom Analytics Dashboard displaying diagnostic metrics by location and test category.

### 2.3 Stakeholders

| Stakeholder       | Role                    | Needs / Expectations                                               | Automation Impact                                                           |
|-------------------|-------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Patients          | End Users               | Easy booking, transparent pricing, timely report delivery          | Self-service portal booking, email notifications with dynamic report links  |
| Lab Technicians   | Operations              | Structured form entry for findings, clear schedule visibility      | Standardized report entry tables, auto-linked patient & appointment context |
| Medical Approvers | Quality Control         | Verification of test results before patient dispatch               | One-click approval workflows and status tracking                            |
| Clinic Managers   | Operations / Management | Operational insights, patient throughput, test volumes             | Interactive Platform Analytics dashboards                                   |
| ServiceNow Admin  | System Admin            | Scalable data schema, automated workflows, reliable email dispatch | Maintainable custom application scope with full audit logging               |

## 3. REQUIREMENT ANALYSIS AND PLANNING

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### 3.1 Execution Roadmap

1. Application Scope & Data Model Definition
2. Diagnostic Test Catalog Configuration
3. Patient Table Schema Setup
4. Appointment Portal & Processing Workflow Design
5. Report Generation Module Implementation
6. Automated Email Notification & Template Scripting
7. Platform Analytics Dashboard Creation
8. End-to-End System Integration Testing & Validation

### 3.2 System Architecture & Data Schema

The application relies on four primary custom tables built within the custom application scope (**x\_1989076\_medica\_0**):

- **x\_1989076\_medica\_0\_diagnostic\_test\_table**: Stores test details, prices, codes, categories, and active status.
- **x\_1989076\_medica\_0\_patient\_table**: Contains registered patient demographics and user references.
- **x\_1989076\_medica\_0\_appointment\_table**: Links patients to diagnostic tests, tracking scheduled dates, slot times, payment status, approval status, and total fees.
- **x\_1989076\_medica\_0\_report\_table**: Captures rich-text clinical findings, report files, verified-by references, and report numbers.

### 3.3 Expected Application Lifecycle

The patient selects a test from the portal catalog, picks a date and time slot, and submits the booking. The system creates an Appointment Record set to Pending Approval. Once approved by administrative staff, a confirmation email is automatically sent to the patient. Following test completion at the center, the lab technician opens the Report Table, enters clinical findings, attaches the test file, and submits it. Saving the report triggers an automated notification containing personalized details (`${patient.first_name}` and `${report_number}`) informing the patient that their report is ready for download.

## 4. DIAGNOSTIC TEST CATALOG & PATIENT MANAGEMENT

### 4.1 Diagnostic Test Configuration

The diagnostic test record defines individual lab tests available within the center. Fields include Test Name, Test Code, Category dropdown, Test Duration, Price (Currency), Active flag, and Description.

The screenshot shows the 'Diagnostic Test Table - New Record' form in ServiceNow. The form is titled 'Diagnostic Test Table - New Record' and has a 'Submit' button at the bottom right. The form fields are:

- Test Name:
- Test Code:
- Category:
- Test Duration:
- Price:
- Active: ☐
- Description:

Figure 1: Diagnostic Test New Record Configuration Form

### 4.2 Diagnostic Test Inventory List

Once populated, the diagnostic test catalog provides a comprehensive view of all active procedures offered by the diagnostic center, including test pricing, codes, and duration.

The screenshot shows the 'Active Diagnostic Test List View' in ServiceNow. The table displays a list of active diagnostic tests with columns for Active, Category, Description, Price, Test Code, Test Duration, and Test Name. The table is filtered by 'Active' and shows 12 rows of data.

| Active | Category   | Description                                 | Price    | Test Code | Test Duration | Test Name                   |
|--------|------------|---------------------------------------------|----------|-----------|---------------|-----------------------------|
| Search | Search     | Search                                      | Search   | Search    | Search        | Search                      |
| true   | Blood Test | Measures different components of blood...   | \$45.00  | CBC-001   | 30            | Complete Blood Count (CBC)  |
| true   | Blood Test | Assesses liver health by measuring enzym... | \$75.00  | LFT-005   | 30            | Liver Function Test (LFT)   |
| true   | Blood Test | Records electrical activity of the heart... | \$55.00  | ECG-009   | 20            | ECG (Electrocardiogram)     |
| true   | Blood Test | Measures cholesterol levels (total, LDL,... | \$60.00  | LIP-002   | 30            | Lipid Profile               |
| true   | Blood Test | Evaluates kidney health through creatini... | \$65.00  | KFT-006   | 30            | Kidney Function Test (KFT)  |
| true   | Blood Test | Highly accurate molecular test for activ... | \$120.00 | COV-010   | 45            | COVID-19 PCR Test           |
| true   | Blood Test | Measures average blood sugar levels over... | \$70.00  | HBA-003   | 30            | HbA1c (Diabetes)            |
| true   | Blood Test | Analyzes urine for infections, kidney di... | \$25.00  | UR-007    | 20            | Urine Routine & Microscopy  |
| true   | Blood Test | Measures Vitamin D levels (25-hydroxyvit... | \$55.00  | VIT-011   | 30            | Vitamin D Test              |
| true   | Blood Test | Evaluates thyroid gland function by meas... | \$95.00  | THY-004   | 30            | Thyroid Function Test (TFT) |

Figure 2: Active Diagnostic Test List View

| Active | Category   | Test Name                   | Test Code | Duration (Min) | Price    |
|--------|------------|-----------------------------|-----------|----------------|----------|
| true   | Blood Test | HbA1c (Diabetes)            | HBA-003   | 30             | \$70.00  |
| true   | Blood Test | Urine Routine & Microscopy  | UR-007    | 20             | \$25.00  |
| true   | Blood Test | Vitamin D Test              | VIT-011   | 30             | \$55.00  |
| true   | Blood Test | Complete Blood Count (CBC)  | CBC-001   | 30             | \$45.00  |
| true   | Blood Test | Thyroid Function Test (TFT) | THY-004   | 30             | \$95.00  |
| true   | Blood Test | Chest X-Ray                 | XRY-008   | 15             | \$80.00  |
| true   | Blood Test | Iron Studies                | IRN-012   | 30             | \$50.00  |
| true   | Blood Test | Liver Function Test (LFT)   | LFT-005   | 30             | \$75.00  |
| true   | Blood Test | ECG (Electrocardiogram)     | ECG-009   | 20             | \$55.00  |
| true   | Blood Test | Lipid Profile               | LIP-002   | 30             | \$60.00  |
| true   | Blood Test | Kidney Function Test (KFT)  | KFT-006   | 30             | \$65.00  |
| true   | Blood Test | COVID-19 PCR Test           | COV-010   | 45             | \$120.00 |

### 4.3 Patient Registration & Data Management

Patient records are captured using the Patient form, which securely logs Patient ID, First Name, Last Name, Phone Number, Email, Gender (Female, Male, Other), Date Of Birth, Address, and the linked SysUser reference.

The screenshot shows a web browser window with the URL `dev440820.service-now.com/now/nav/ui/classic/params/target/x_1989076_medica_0_patient_table.do%3Fsys_id%3D-1`. The ServiceNow interface displays the 'Patient Table - New Record' form. The form fields are as follows:

- Patient ID**: A text input field with a green border.
- First Name**: A text input field.
- Last Name**: A text input field.
- Phone Number**: A text input field with a light blue background.
- Email**: A text input field.
- Gender**: A dropdown menu currently set to 'Female'.
- Date Of Birth**: A date picker field.
- Address**: A text input field.
- User**: A text input field with a search icon on the right.

A 'Submit' button is located at the bottom left of the form area.

Figure 3: Patient Registration Form Configuration

## 5. APPOINTMENT SCHEDULING & WORKFLOW MANAGEMENT

### 5.1 Booking & Scheduling Interface

Patients can select their required test, schedule a date/time slot, and submit the booking through a guided multi-step wizard. The portal homepage highlights available tests and average report turnaround time, giving patients a transparent, self-service entry point into the diagnostic center.

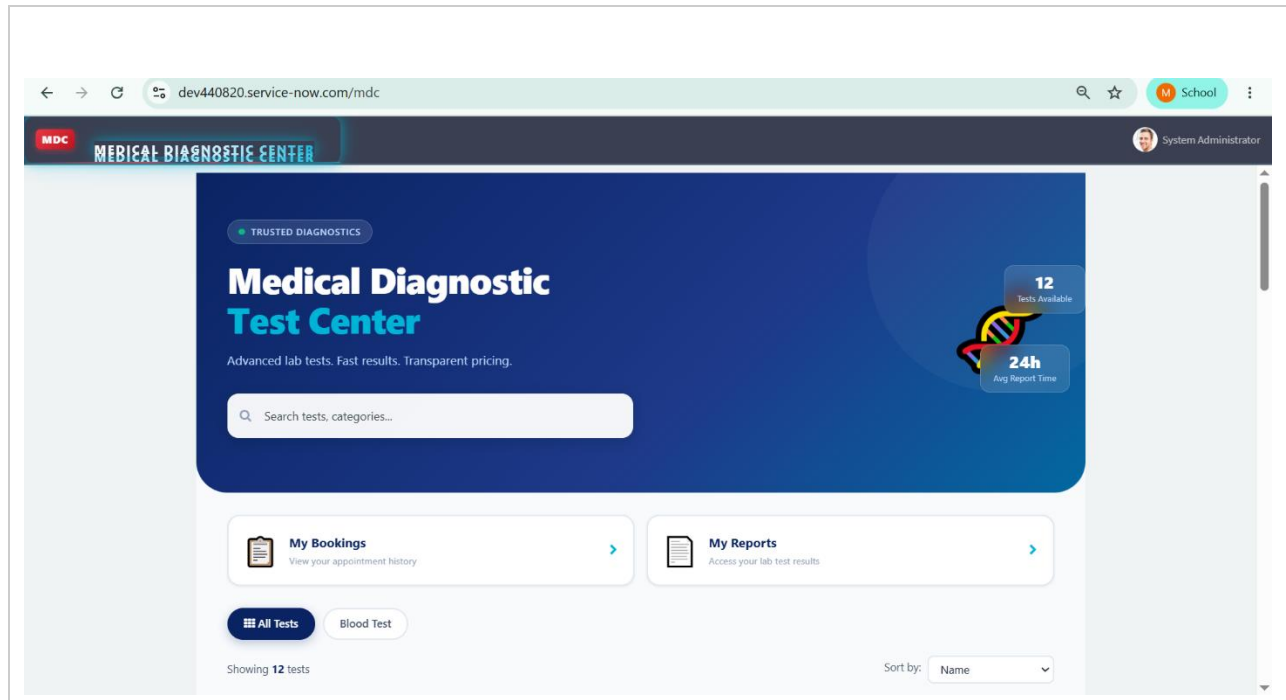


Figure 4: Self-Service Patient Portal Homepage

From the homepage, patients browse the full catalog of active diagnostic tests, filter by category, and initiate a booking directly against the desired test.



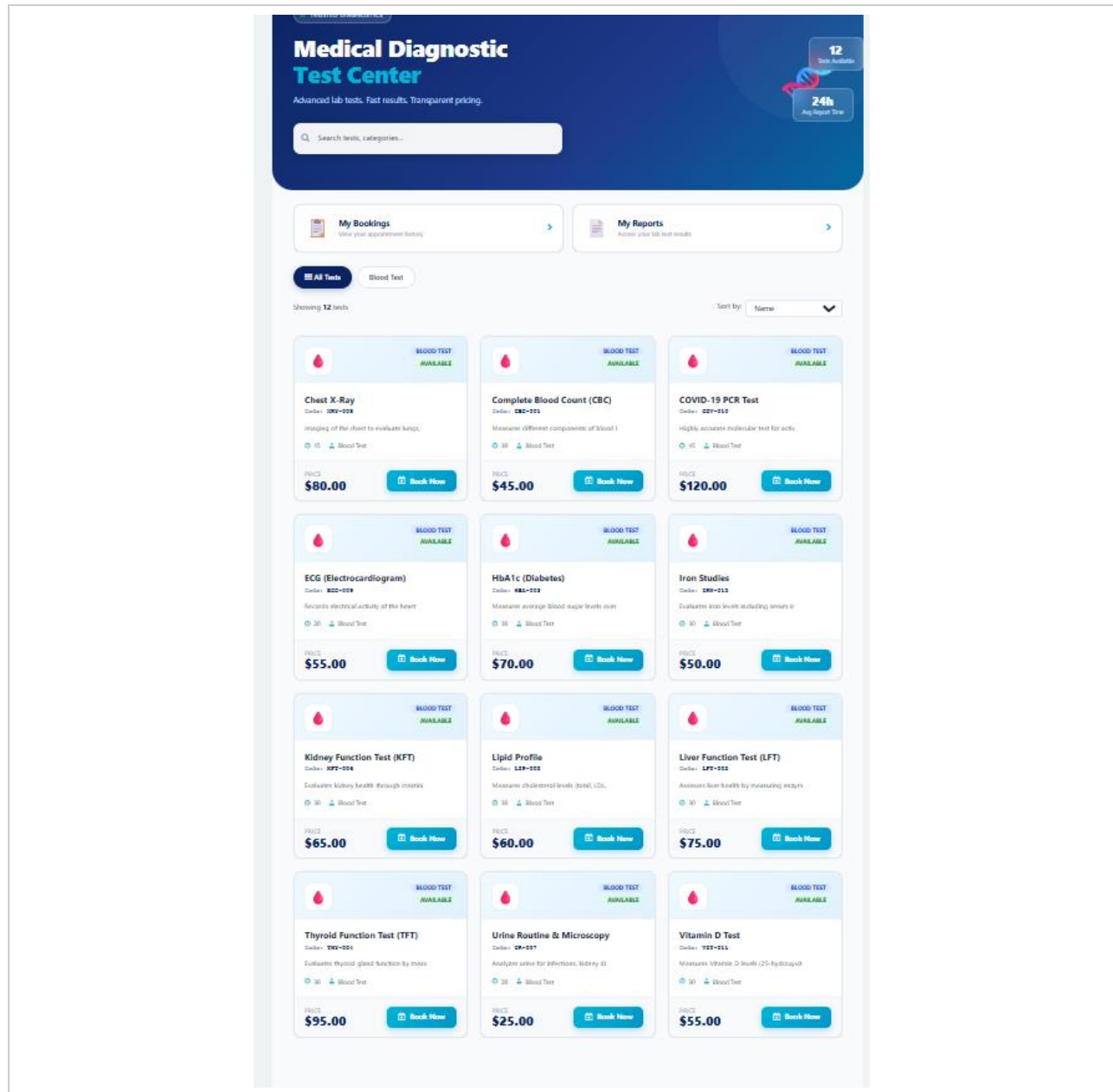


Figure 5: Diagnostic Test Catalog — Patient Portal View

Selecting a test opens the guided booking wizard. Step one collects the patient's personal information — name, contact details, date of birth, gender, and address — before the patient proceeds to schedule and pay for the appointment.

1

Details

2

Schedule

3

Payment

4

Done



**Chest X-Ray**  
Blood Test · Code: XRY-008 · Duration: 15

**\$80.00**  
[Change Test](#)



**Patient Information**  
Fill in your personal details to proceed.

FIRST NAME \*

LAST NAME \*

EMAIL ADDRESS \*

PHONE NUMBER \*

DATE OF BIRTH \*  


GENDER \*  

-- Select Gender --



ADDRESS  


[← Back to Catalog](#)

[Continue →](#)

Figure 6: Patient Information Entry — Booking Wizard, Step 1

Upon completing the scheduling and payment steps, the screen provides instant confirmation of the appointment ID (APPT-2026-24623), payment mode, total cost, and pending approval status.

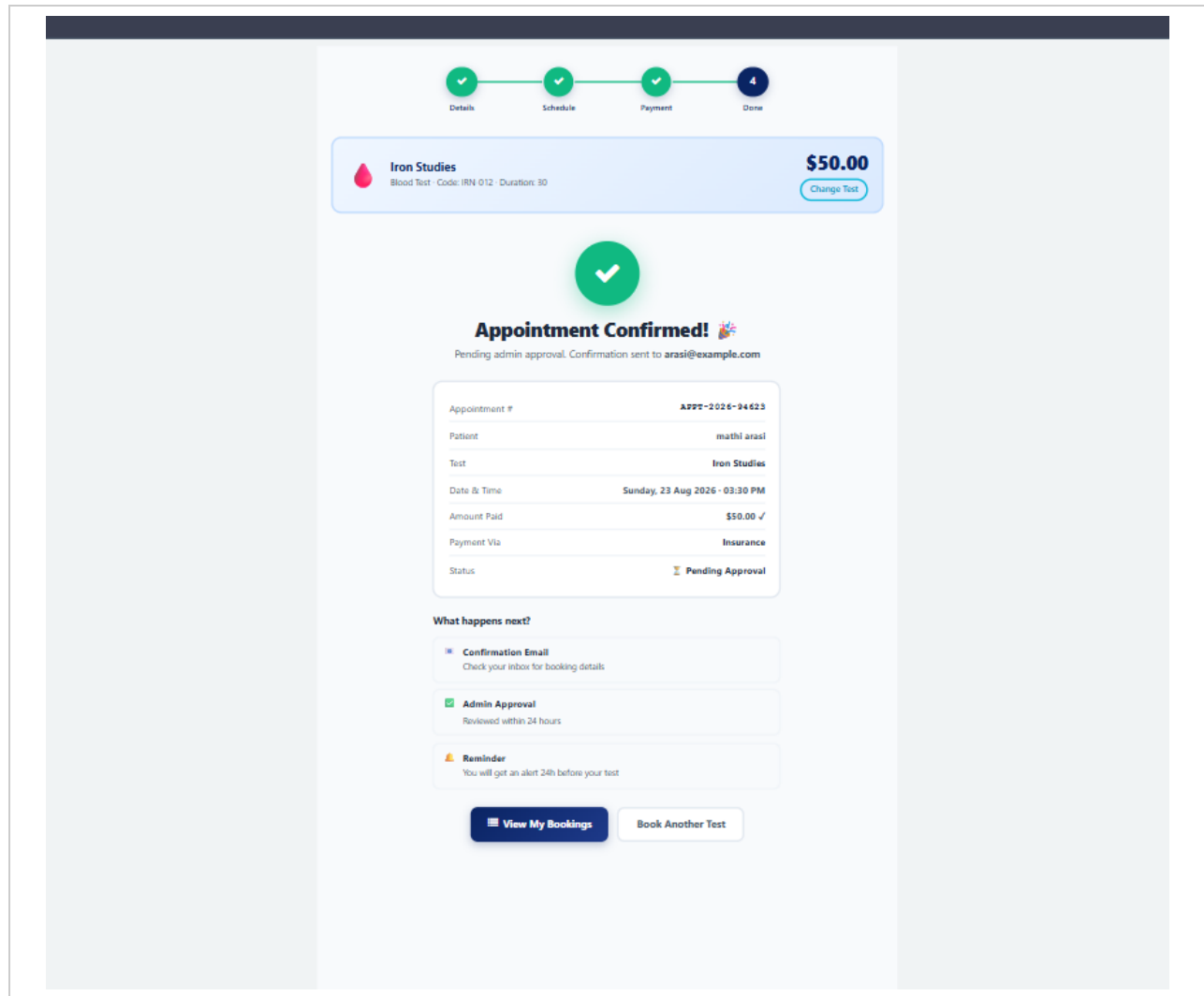


Figure 7: Patient Appointment Confirmation Interface

## 5.2 Appointment Table Schema & Record Structure

The back-end Appointment Table (x\_1989081\_medica\_0\_appointment\_table) tracks all scheduled bookings. Administrators can view reference links to Patient and Diagnostic Test records, alongside financial and approval metadata.

dev440820.service-now.com/now/nav/ui/classic/params/target/x\_1989076\_medica\_0\_appointment\_table.do%3Fsys\_id%3D-1

servicenow All Favorites History Workspaces Admin APPOINTMENT TABLE - New Record Search

APPOINTMENT TABLE New record

Appointment Number

Patient

Status -- None --

Approval Status -- None --

Email

Approval Date

Test

Slot Time

Total Amount \$ 0.00

Rejection Reason

Confirmation Sent

Created By

Approved By

Notes

Payment Status -- None --

Scheduled Date

Refund Amount \$ 0.00

Submit

Figure 8: Appointment Table Form Interface

The list view below shows the aggregate of appointment records, including approval status, confirmation flags, payment status, and the associated patient reference.

dev440820.service-now.com/now/nav/ui/classic/params/target/x\_1989076\_medica\_0\_appointment\_table\_list.do

servicenow All Favorites History Workspaces Admin APPOINTMENT TABLES Search

APPOINTMENT TABLES Appointment Number Search Actions on selected rows... New

| Appointment Number | Approval Date       | Approval Status | Approved By          | Confirmation Sent | Created By           | Email               | Notes                        | Patient             | Payment Status |
|--------------------|---------------------|-----------------|----------------------|-------------------|----------------------|---------------------|------------------------------|---------------------|----------------|
| APPT-2026-63326    | (empty)             | Approved        | System Administrator | true              | System Administrator | testuser@gmail.com  | Approved by system admin     | 2026-08-24 23:05:54 | paid           |
| APPT-2026-82720    | 2026-08-27 00:48:17 | Approved        | (empty)              | true              | System Administrator | jphndoo@gmail.com   |                              | 2026-08-24 23:05:36 | paid           |
| APPT-2026-77498    | 2026-08-27 00:54:20 | Approved        | (empty)              | true              | System Administrator | premkumar@gmail.com |                              | 2026-08-26 00:53:18 | paid           |
| APPT-2026-49505    | 2026-08-26 23:12:53 | Approved        | (empty)              | true              | System Administrator | testuser@gmail.com  |                              | 2026-08-24 23:05:54 | paid           |
| APPT-2026-46155    | 2026-08-27 02:17:16 | Approved        | System Administrator | true              | System Administrator | admin@example.com   | Approved by the system admin | 2026-08-25 01:54:00 | paid           |
| APPT-2026-37653    | 2026-08-26 01:07:20 | Approved        | (empty)              | true              | System Administrator | tharun@gmail.com    |                              | 2026-08-26 01:06:24 | paid           |
| APPT-2026-26252    | (empty)             | requested       | (empty)              | true              | System Administrator | testuser@gmail.com  |                              | 2026-08-24 23:05:54 | paid           |
| APPT-2026-12072    | (empty)             | requested       | (empty)              | true              | System Administrator | testuser@gmail.com  |                              | 2026-08-24 23:05:54 | paid           |
| APPT-2026-10846    | 2026-08-26 01:01:36 | Approved        | (empty)              | true              | System Administrator | tharan@gmail.com    |                              | 2026-08-26 01:00:35 | paid           |

Figure 9: Appointment Table List View

## Appointment Record Field Map

- **Appointment Number:** Auto-generated unique string identifier (e.g., APPT-2026-24623).
- **Patient:** Reference field pointing to x\_1989081\_medica\_0\_patient.
- **Status:** Choice dropdown (Pending, Confirmed, Completed, Cancelled).
- **Approval Status:** Choice dropdown (Pending Approval, Approved, Rejected).
- **Scheduled Date & Slot Time:** Date/Time picker capturing the exact slot reserved.
- **Test:** Reference field pointing to x\_1989081\_medica\_0\_diagnostic\_test.
- **Total Amount & Refund Amount:** Currency fields tracking financial transactions.
- **Payment Status:** Dropdown reflecting payment processing (Pending, Paid, Refunded).

- **Confirmation Sent:** Checkbox flag set automatically by notification scripts.

## 6. REPORT GENERATION & NOTIFICATION INTEGRATION

### 6.1 Diagnostic Report Schema

Upon completing a lab test, technicians record findings in the Report Table (x\_1989081\_medica\_0\_report\_table). This table features an HTML editor for detailed clinical notes, file attachment fields for uploaded PDF lab results, and reference fields linking back to the specific Patient and Appointment.

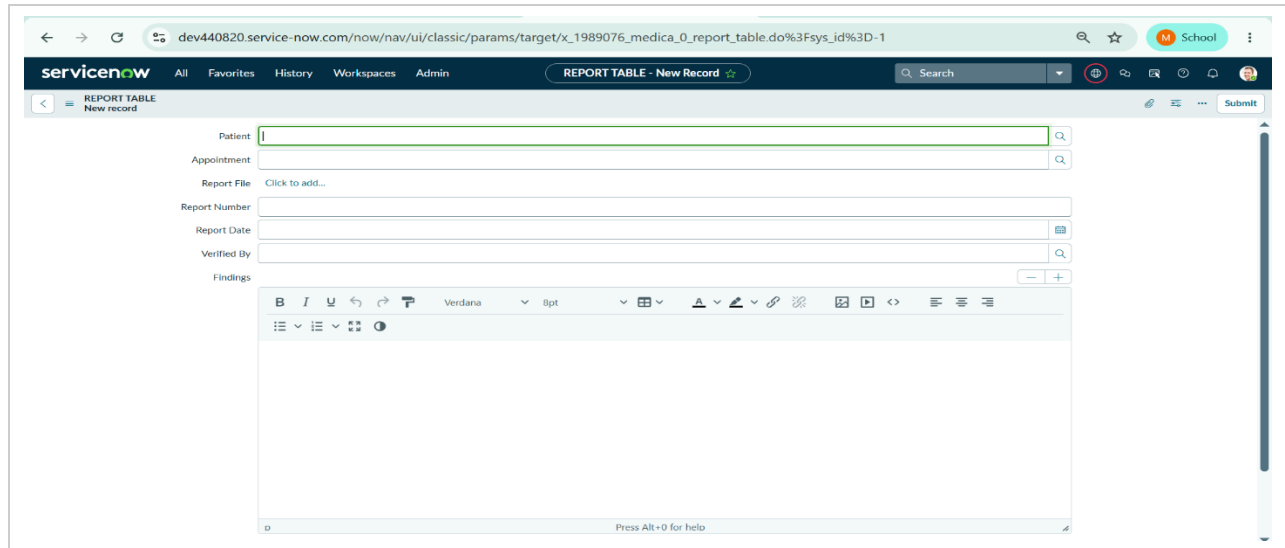


Figure 10: Diagnostic Report Form Interface

### 6.2 Automated Email Notification System

When a diagnostic report is verified and saved, ServiceNow triggers an automated email notification (reportsNotifications). The subject and body dynamically interpolate record attributes using script variables such as `${patient.first_name}` and `${report_number}`.

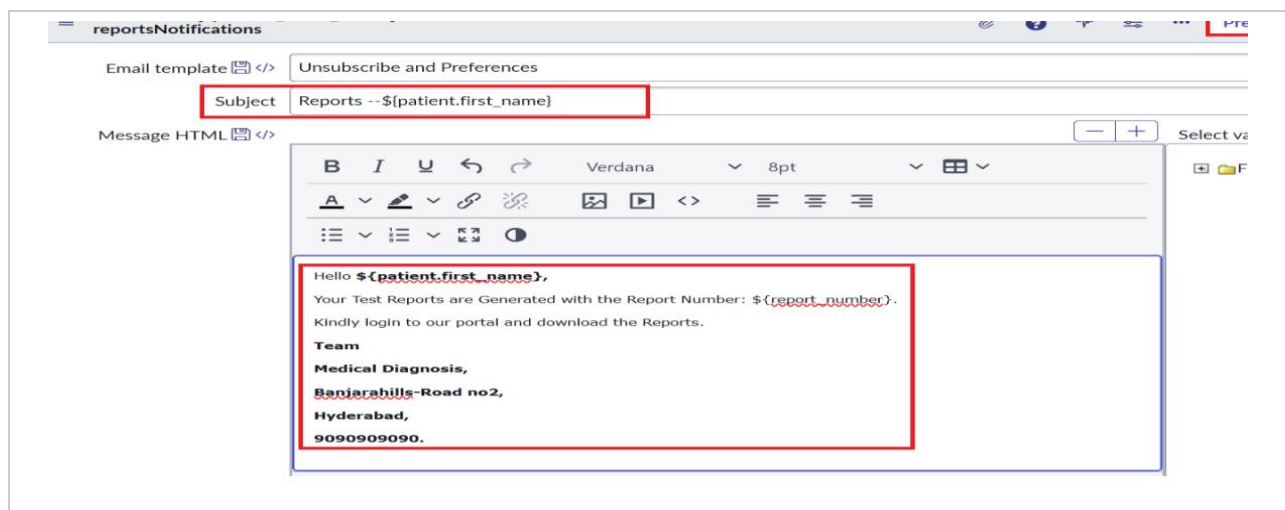


Figure 11: Report Email Notification Template Configuration

### Email Notification Content

- **Subject:** Reports --\${patient.first\_name}

#### Body:

*Hello \${patient.first\_name},*

*Your Test Reports are Generated with the Report Number: \${report\_number}.*

*Kindly login to our portal and download the Reports.*

**Team**

**Medical Diagnosis,**

**Banjarahills-Road no2,**

**Hyderabad,**

**9090909090.**

## 6.3 Email Dispatch Logs & Restocking Alerts

The system monitors outgoing mail using ServiceNow's native Email Engine (sys\_email). System administrators can audit generated notification payloads, verify recipient addresses, review delivery statuses, and monitor system events (such as inventory restocking alerts for diagnostic supplies).

The screenshot displays the ServiceNow user interface. At the top, the navigation bar includes 'All', 'Favorites', 'History', and a search bar. The main header shows 'Email - Restocking Request For Fujit...'. A modal window titled 'Preview Email' is open, showing the email content:

Dear Aileen Mottern,

The current stock for Fujitsu 1TB Hybrid Solid State Drive has fallen under the threshold value of 10.

Please order 50 more for Santa Monica Warehouse, so that it is above the threshold.

[Unsubscribe](#) | [Notification Preferences](#)

Ref:MSG0000157\_4Z73EEEn72f7Jlow9jkyp

&subject=Unsubscribe+from+%22Asset+Restocking%22>  
X-ServiceNow-SysEmail-Version:2

Below the preview, there are 'Update' and 'Delete' buttons. Under 'Related Links', there is a link to 'review Email'. Below that, there are tabs for 'Email Log (1)' and 'Email Attachments'. The 'Email Log' tab is active, showing a table with columns: 'Created', 'Level', 'Message', and 'Source'.

| Created             | Level       | Message                                                                                                                                                                                                                     | Source                                 |
|---------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 2026-08-25 01:00:06 | Information | Notification 'Asset Restocking' (cb3d87a2c3131000b959fd251eba8f81) included recipients via notification's 'Users/Groups in fields' (e.g., watchlist, assigned_to, etc): 'Aileen Mottern' (71826bf03710200044e0bfc8bcbe5d3b) | EMAIL:31d34963833e43109b27f4b6feaad37e |

Figure 12: Email Preview and System Audit Log View

## 7. ANALYTICS & HEALTH REPORTS DASHBOARD

### 7.1 Platform Analytics & Reporting

To provide management with visibility into diagnostic center operations, a custom Health Reports Dashboard was configured using the ServiceNow Platform Analytics workspace.

### 7.2 Dashboard Visualization

The dashboard displays real-time metrics, such as report distribution across different regional branches/locations (e.g., Cuddalore vs. MYD centers).

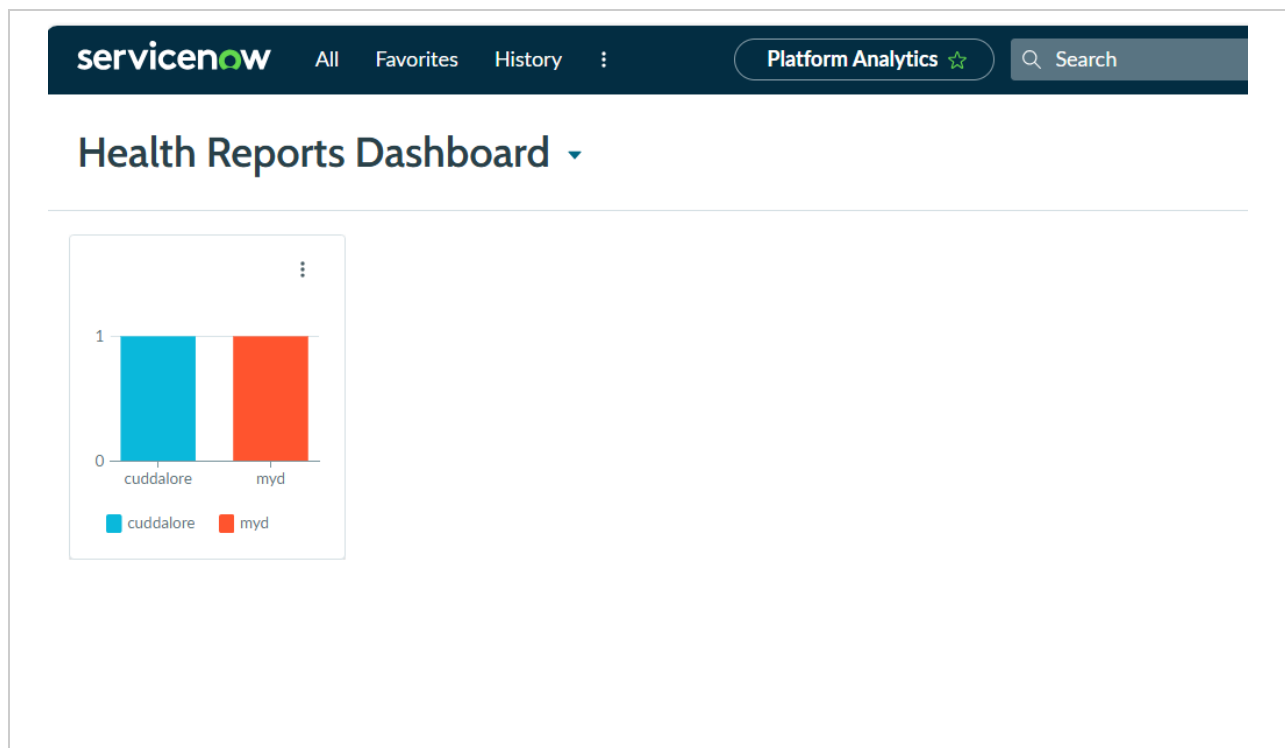


Figure 13: Health Reports Dashboard in Platform Analytics



## 8. FINAL VALIDATION, TESTING AND CONCLUSION

### 8.1 Final Validation

The final phase verifies that all operational steps — from patient registration to dashboard metrics — function seamlessly:

- **Patient Registration:** Successfully creates user profiles with linked contact details.
- **Test Catalog Access:** Tests are correctly listed with active pricing and descriptions.
- **Appointment Scheduling:** Appointments generate auto-numbered records and trigger approval workflows.
- **Report Generation:** Lab reports save HTML findings and attach diagnostic PDFs.
- **Email Notification Dispatch:** Emails render dynamic fields (`${patient.first_name}`, `${report_number}`) and log correctly in `sys_email`.
- **Platform Analytics:** Real-time dashboards accurately reflect location-based test counts.

### 8.2 End-to-End Testing & Deployment Validation

| Phase | Main Activity                        | Expected Result                                      | Status |
|-------|--------------------------------------|------------------------------------------------------|--------|
| 1     | Requirement Analysis & Data Modeling | Custom scoped tables defined (x_1989081_medica_0)    | PASSED |
| 2     | Diagnostic Test Catalog Setup        | Test catalog populated with active tests and prices  | PASSED |
| 3     | Patient Registration                 | Patient demographic data accurately mapped           | PASSED |
| 4     | Appointment Scheduling               | Portal wizard schedules appointment with ID tracking | PASSED |
| 5     | Approval & Billing                   | Appointment status updates to pending/approved       | PASSED |
| 6     | Report Generation                    | Clinical findings logged via rich-text editor        | PASSED |
| 7     | Notification Triggering              | Email generated with dynamic script interpolation    | PASSED |
| 8     | Audit & Log Verification             | Outgoing emails verified in ServiceNow Email Logs    | PASSED |
| 9     | Analytics Dashboard                  | Real-time health report charts active in workspace   | PASSED |

### 8.3 Project Outcome

The Medical Diagnostic Center application demonstrates how ServiceNow custom application development can automate healthcare operations. By integrating table design, dynamic portal workflows, rich-text report generation, automated email notifications, and Platform Analytics dashboards, the solution minimizes administrative overhead and improves patient satisfaction.

## **8.4 Conclusion**

The successful implementation of the Medical Diagnostic Center platform provides a complete solution for managing diagnostic workflows. The application accelerates report delivery, eliminates scheduling errors, maintains transparent transaction logs, and delivers actionable insights to healthcare administrators.